

InfraGuard Project - Team Contribution & Supporting Document

PROJECT: InfraGuard - Infrastructure Inventory Reconciliation System

Project Overview

InfraGuard is a full-stack Infrastructure Inventory Reconciliation System built using FastAPI, SQLite, AI integrations, PDF reporting, and a web dashboard. The system compares CMDB inventory data with actual infrastructure scan data, identifies discrepancies, evaluates risks, and generates actionable reports.

Team Contribution Distribution (4 Members):

Member-1 (Sukant.B) - Backend & Database Development

- Designed FastAPI backend architecture.
- Developed API endpoints for upload, analysis, reporting, chat, and orchestration.
- Configured SQLite database and asynchronous database operations.
- Managed application startup, routing, and backend integration.

Member-2 (Karthikeyan.S) - Data Processing & Reconciliation Engine

- Implemented data ingestion and validation services.
- Developed normalization logic for hostnames, IP addresses, RAM, OS versions, and statuses.
- Built reconciliation engine for asset matching and discrepancy detection.
- Implemented duplicate, missing, and untracked asset identification.

Member-3 (Dharineesh.S) - AI & Risk Analysis Module

- Integrated OpenAI, Gemini, and vector search capabilities.

- Developed chatbot and AI-assisted discrepancy summarization.
- Implemented risk assessment and severity classification.
- Generated recommendations and investigation workflows.

Member-4 (Selva vikash.P) - Frontend, Reporting & Deployment

- Developed dashboard using HTML, CSS, and JavaScript.
- Created charts, tables, and user interaction features.
- Implemented PDF report generation using ReportLab.
- Managed testing, deployment, documentation, and project presentation.

Tools & Technologies

- FastAPI
- Python
- SQLite
- HTML, CSS, JavaScript
- ReportLab
- OpenAI API
- Google Gemini API
- MongoDB Atlas / ChromaDB
- Git & GitHub

Project Outcomes

- Automated infrastructure inventory reconciliation.
- Improved asset visibility and discrepancy detection.
- Generated risk-based recommendations.
- Reduced manual auditing effort through automation and AI assistance.